

Exploration catalogue: every trained configuration of the CIFAR-10 study

Companion record to *Continuation Methods for Training Convolutional Neural Networks*

Generated by `paper/tools/build_catalogue.py` from the run records

This catalogue lists every configuration trained during the CIFAR-10 exploration that precedes the selection of the three procedures, grouped by experiment. It is a traceability record, not a ranking: the experiments differ in data, budget, evaluation split and inference state, and are therefore not comparable across groups. Accuracy is the final recorded evaluation of each run, averaged over the numerically valid seeds, with no best-epoch selection; SD is the sample standard deviation over those seeds and is absent when a single seed is valid. Seeds are reported as valid/attempted, so diverged runs remain visible. Wall time is the mean over valid runs, on the timing scope stated in each group header.

Every value here was recomputed from the per-run `summary.json` records; see `CHANGE_LOG.md`.

Configuration	Acc. (%)	SD	Seeds	Wall (s)
Promising methods in one batch, evaluated every epoch				
Promising methods in one batch, evaluated every epoch (unified_selected; owner Max). Conditions: reference recipe; evaluation after every epoch; per-epoch mean training loss recorded. Evaluation split: test. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel..				
Baseline: no blur, full 32x32 throughout	75.43	0.76	3/3	505
Blur after every ReLU, annealed to zero	79.91	0.27	3/3	624
Blur every conv layer, annealed to zero	78.37	0.77	3/3	737
Shrink block 0 with max-pool, 16 to 24 to 32	80.24	0.48	3/3	483
Shrink block 0, plus blur after every ReLU	80.80	0.41	3/3	623
Shrink block 0, plus blur scaled by receptive field after every ReLU	81.11	0.37	3/3	620
Shrink block 1 with max-pool, 16 to 24 to 32	80.37	0.61	3/3	499
Shrink block 1, plus blur after every ReLU	81.00	0.43	3/3	631
Shrink block 1, plus blur every conv layer	81.51	0.22	3/3	733
Shrink block 1, plus blur scaled by receptive field after every ReLU	81.05	0.57	3/3	624
Shrink block 1, plus blur scaled by sqrt of map size after every ReLU	80.10	0.20	3/3	636
Shrink block 2 with max-pool, 16 to 24 to 32	80.12	0.45	3/3	499
Shrink block 2, plus blur after every ReLU	80.95	0.25	3/3	620
Shrink block 2, plus blur scaled by receptive field after every ReLU	80.94	0.17	3/3	621
Shrink the input image, 16 to 24 to 32	79.28	0.53	3/3	482
Shrink the input image, plus blur after every ReLU	80.92	0.38	3/3	619
Resolution-only benchmark: 7 operators x 5 sites x 4 schedules, all Gaussian disabled				
Resolution-only benchmark: 7 operators x 5 sites x 4 schedules, all Gaussian disabled (resbench_resolution_only; owner Max). Conditions: reference recipe; evaluation every 2 epochs; fixed16 / fixed24 controls report their reduced-resolution path. Evaluation split: test. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel..				
Baseline: no blur, full 32x32 throughout (resolution batch)	75.53	0.36	3/3	428
Shrink block 0 with max-pool, 16 to 24 to 32	80.08	0.61	3/3	438
Shrink block 1 with bilinear, 16 to 24 to 32	80.23	0.38	3/3	436
Shrink block 1 with least-squares reconstruction, 16 to 24 to 32	80.11	0.52	3/3	510
Shrink block 1 with max-pool then anti-alias blur, 16 to 24 to 32	80.43	0.75	3/3	519
Shrink block 1 with max-pool then anti-alias blur, 16 to 24 to 32, longer at low res	80.20	0.52	3/3	446
Shrink block 1 with max-pool then anti-alias blur, 24 to 16 to 32 (order control)	80.46	0.44	3/3	448

continued on next page

Configuration	Acc. (%)	SD	Seeds	Wall (s)
Shrink block 1 with max-pool then anti-alias blur, 24 to 32 only	80.32	0.22	3/3	499
Shrink block 1 with max-pool, 16 to 24 to 32	80.60	0.20	3/3	433
Shrink block 1 with max-pool, 16 to 24 to 32, longer at low res	80.07	0.53	3/3	436
Shrink block 1 with max-pool, 24 to 16 to 32 (order control)	80.21	0.30	3/3	438
Shrink block 1 with max-pool, 24 to 32 only	79.78	0.32	3/3	430
Shrink block 1 with max-pool, fixed 16x16, never restored (fixed-resolution control: trained and evaluated at 16x16 after block/site reduction, final metric is that reduced path, the 32x32 path is a diagnostic)	73.13	0.54	3/3	419
Shrink block 1 with max-pool, fixed 24x24, never restored (fixed-resolution control: trained and evaluated at 24x24, final metric is that reduced path)	76.15	0.49	3/3	436
Shrink block 1 with perceptual (SSIM-style), 16 to 24 to 32 (diverged)	—	—	0/3	—
Shrink block 1 with smoothness-optimal reconstruction, 16 to 24 to 32	79.85	0.76	3/3	507
Shrink block 1 with smoothness-optimal reconstruction, 16 to 24 to 32, longer at low res	79.55	0.66	3/3	564
Shrink block 1 with smoothness-optimal reconstruction, 24 to 16 to 32 (order control)	79.50	0.41	3/3	527
Shrink block 1 with smoothness-optimal reconstruction, 24 to 32 only	78.90	0.11	3/3	548
Shrink block 1 with softmax-weighted pooling, 16 to 24 to 32	80.34	0.48	3/3	482
Shrink block 2 with max-pool, 16 to 24 to 32	80.24	0.56	3/3	428
Shrink the first conv layer with max-pool, 16 to 24 to 32	79.61	0.82	3/3	409
Shrink the input image with bilinear, 16 to 24 to 32	79.56	0.68	3/3	427
Shrink the input image with least-squares reconstruction, 16 to 24 to 32	79.64	0.33	3/3	537
Shrink the input image with max-pool then anti-alias blur, 16 to 24 to 32	79.17	0.43	3/3	505
Shrink the input image with max-pool, 16 to 24 to 32	79.23	0.48	3/3	433
Shrink the input image with perceptual (SSIM-style), 16 to 24 to 32	79.98	0.40	3/3	462
Shrink the input image with smoothness-optimal reconstruction, 16 to 24 to 32	79.76	0.65	3/3	560
Shrink the input image with softmax-weighted pooling, 16 to 24 to 32	79.28	0.46	3/3	456
21-configuration grid: resolution x blur x site				
21-configuration grid: resolution x blur x site (campaign_grid21; owner Max). Conditions: reference recipe; evaluation every 2 epochs. Evaluation split: test. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel..				
Baseline: no blur, full 32x32 throughout	75.62	0.49	3/3	489
Blend each layer between itself and its blur	78.18	0.36	3/3	789
Blur every conv layer, annealed to zero	78.69	0.68	3/3	696
Blur every conv layer, annealed to zero, first 7 layers only	75.24	0.74	3/3	593
Blur every conv layer, geometric decay	78.28	0.50	3/3	703
Shrink the first conv layer (bilinear), 16 to 24 to 32	78.91	0.37	3/3	484
Shrink the first conv layer (bilinear), 16 to 24 to 32, plus blur every conv layer, annealed to zero	79.30	0.62	3/3	750
Shrink the first conv layer (max-pool), 16 to 24 to 32	79.97	0.58	3/3	491
Shrink the first conv layer (max-pool), 16 to 24 to 32, plus blur every conv layer, annealed to zero	79.81	0.79	3/3	747
Shrink the input image (bilinear), 16 to 24 to 32	79.48	0.54	3/3	492
Shrink the input image (bilinear), 16 to 24 to 32, plus blend each layer between itself and its blur	79.51	0.80	3/3	783
Shrink the input image (bilinear), 16 to 24 to 32, plus blur every conv layer, annealed to zero	80.71	0.65	3/3	700
Shrink the input image (bilinear), 16 to 24 to 32, plus blur every conv layer, annealed to zero, first 7 layers only	79.27	0.57	3/3	596
Shrink the input image (bilinear), 16 to 24 to 32, plus blur every conv layer, geometric decay	79.91	0.92	3/3	709
Shrink the input image (bilinear), 24 to 16 to 32 (order control)	79.23	0.40	3/3	467

continued on next page

Configuration	Acc. (%)	SD	Seeds	Wall (s)
Shrink the input image (bilinear), 24 to 16 to 32 (order control), plus blur every conv layer, annealed to zero	79.33	0.38	3/3	757
Shrink the input image (bilinear), 24 to 32 only	78.56	0.34	3/3	486
Shrink the input image (bilinear), 24 to 32 only, plus blur every conv layer, annealed to zero	79.77	0.72	3/3	694
Shrink the input image (bilinear), 24 to 32 only, plus blur every conv layer, geometric decay	79.68	0.57	3/3	704
Shrink the input image (max-pool), 16 to 24 to 32	79.80	0.50	3/3	489
Shrink the input image (max-pool), 16 to 24 to 32, plus blur every conv layer, annealed to zero	80.34	0.76	3/3	738
Anti-aliasing ablation: placement, masks, constant sigma, BlurPool, internal reductions, per-layer profiles				
Anti-aliasing ablation: placement, masks, constant sigma, BlurPool, internal reductions, per-layer profiles (ablation_aa; owner Idriss). Conditions: reference recipe; own pinned initial weights (r20bn-ablation-assets); evaluation every 2 epochs. Evaluation split: test. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel..				
Baseline: no blur, full 32x32 throughout (ablation batch)	75.06	—	1/1	423
Blur after every BatchNorm, annealed to zero	78.39	—	1/1	673
Blur after every ReLU (10 places), annealed to zero	80.08	0.21	3/3	538
Blur every conv layer, annealed to zero	78.18	—	1/1	648
Blur only the 2 layers feeding a downsample	77.55	—	1/1	436
Blur rising with depth, after every ReLU (direction control)	77.01	0.38	3/3	585
Blur scaled by map size, after every ReLU	78.48	0.33	3/3	550
Blur scaled by receptive field, after every ReLU	79.59	0.10	3/3	532
Blur scaled by sqrt of map size, after every ReLU	79.55	0.46	3/3	579
Blur the 17 layers away from a downsample	78.52	—	1/1	605
Fixed anti-alias blur before each downsample (control) (fixed BlurPool architecture kept at inference: its target path is not the plain ResNet-20)	78.51	—	1/1	460
Fixed anti-alias blur before each downsample, plus annealed blur everywhere (fixed BlurPool architecture kept at inference: its target path is not the plain ResNet-20)	79.19	—	1/1	698
Fixed blur 0.30, never annealed (control) (fixed-filter control: constant sigma, never annealed, final metric is the filtered (current) path)	74.97	—	1/1	755
Fixed blur 0.50, never annealed (control) (fixed-filter control: constant sigma, never annealed, final metric is the filtered (current) path)	78.69	—	1/1	738
Fixed blur 0.80, never annealed (control) (fixed-filter control: constant sigma, never annealed, final metric is the filtered (current) path)	75.02	—	1/1	736
Fixed blur 1.00, never annealed (control) (fixed-filter control: constant sigma, never annealed, final metric is the filtered (current) path)	71.15	—	1/1	749
Fixed blur after every ReLU, never annealed (control) (fixed-filter control: constant post-ReLU sigma, final metric is the filtered (current) path)	79.78	—	1/1	557
Shrink after the first ReLU, plus blur after every ReLU	79.65	—	1/1	536
Shrink block 0 with max-pool, 16 to 24 to 32	80.44	0.43	3/3	455
Shrink block 0, plus blur after every ReLU	80.53	—	1/1	545
Shrink block 1 with max-pool, 16 to 24 to 32	80.59	0.41	3/3	447
Shrink block 1, plus blur after every ReLU	80.08	—	1/1	519
Shrink block 2 with max-pool, 16 to 24 to 32	80.09	0.17	3/3	447
Shrink block 2, plus blur after every ReLU	80.99	0.41	3/3	574
Shrink block 2, plus blur rising with depth after every ReLU (direction control)	80.18	0.46	3/3	583
Shrink block 2, plus blur scaled by map size after every ReLU	79.61	0.44	3/3	633
Shrink block 2, plus blur scaled by receptive field after every ReLU	81.14	0.22	3/3	594
Shrink block 2, plus blur scaled by sqrt of map size after every ReLU	80.28	0.24	3/3	551
Shrink the first conv layer, 16 to 24 to 32	79.95	—	1/1	437
Shrink the input image, 16 to 24 to 32	78.92	—	1/1	443

continued on next page

Configuration	Acc. (%)	SD	Seeds	Wall (s)
Shrink the input image, 16 to 24 to 32, plus blur after every ReLU	80.14	—	1/1	542
Shrink the input image, 16 to 24 to 32, plus blur every conv layer	79.15	—	1/1	604
Per-layer sigma profiles (rho), adaptive controller, sigma0 sweep				
Per-layer sigma profiles (rho), adaptive controller, sigma0 sweep (per_layer_sigma; owner Aubin). Conditions: own self-paired assets per seed (different digest scheme); per_layer_cpu is a 6,000/2,000-image CPU pilot of 12 epochs. Evaluation split: test. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel..				
Baseline: no blur, full 32x32 throughout (per-layer batch) [5000 train / 2000 eval, 4 epochs]	28.40	—	1/1	—
Baseline: no blur, full 32x32 throughout (per-layer batch) [6000 train / 2000 eval, 12 epochs]	47.65	—	1/1	—
Baseline: no blur, full 32x32 throughout (per-layer batch) [launch per-layer-adaptive-fix-20260910-130520]	73.91	0.62	3/3	—
Baseline: no blur, full 32x32 throughout (per-layer batch) [launch per-layer-sigma-20260910-094757]	74.42	0.29	3/3	—
Baseline: no blur, full 32x32 throughout (per-layer batch), sigma0 = 1	74.13	0.73	3/3	—
Per-layer blur strength chosen by the network [5000 train / 2000 eval, 4 epochs]	31.65	—	1/1	—
Per-layer blur strength chosen by the network [launch per-layer-adaptive-fix-20260910-130520]	77.35	0.49	3/3	—
Per-layer blur strength chosen by the network [launch per-layer-sigma-20260910-094757]	77.50	0.61	3/3	—
Per-layer blur strength, exponent 0.5	74.65	0.67	3/3	—
Per-layer blur strength, exponent 0.5 [6000 train / 2000 eval, 12 epochs]	45.80	—	1/1	—
Per-layer blur strength, exponent 1 [6000 train / 2000 eval, 12 epochs]	49.00	—	1/1	—
Per-layer blur strength, exponent 1 [launch per-layer-adaptive-fix-20260910-130520]	77.66	0.68	3/3	—
Per-layer blur strength, exponent 1 [launch per-layer-sigma-20260910-094757]	77.65	0.74	3/3	—
Per-layer blur strength, exponent 1, sigma0 = 0.25	73.81	0.96	3/3	—
Per-layer blur strength, exponent 1, sigma0 = 0.5	76.41	0.86	3/3	—
Per-layer blur strength, exponent 1, sigma0 = 1	77.84	0.63	3/3	—
Per-layer blur strength, exponent 1, sigma0 = 1.5	74.55	0.38	3/3	—
Per-layer blur strength, exponent 1, sigma0 = 2	70.59	0.71	3/3	—
Per-layer blur strength, exponent 2	76.65	0.21	3/3	—
Per-layer blur strength, exponent 2 [5000 train / 2000 eval, 4 epochs]	31.80	—	1/1	—
Per-layer blur strength, exponent 2 [6000 train / 2000 eval, 12 epochs]	50.10	—	1/1	—
Data-triggered schedules, gradient-norm calibration, dwell allocation, long-horizon runs				
Data-triggered schedules, gradient-norm calibration, dwell allocation, long-horizon runs (adaptive_continuation; owner Aubin). Conditions: reference recipe, mostly seed 0 only; long-conv runs use 120 epochs and are not comparable on accuracy. Evaluation split: test. Timing: wall_seconds from model construction to summary write, including evaluations and checkpoint writes; eval_seconds counts snapshot evaluation only; train_seconds = wall - eval (so it includes data indexing and checkpoint I/O). One T4, two cells in parallel on a two-GPU kernel..				
Blur every conv layer, annealed (instrumented rerun) [launch gradnorm-cal-20260910-105845]	77.97	—	1/1	637
Blur every conv layer, annealed (instrumented rerun) [launch gradnorm-cal2-20260910-111547]	77.94	—	1/1	719
Blur every conv layer, annealed (instrumented rerun) [launch signals-cal3-20260910-113759]	78.33	—	1/1	721
Blur schedule chosen by a transfer-gap trigger (mis-set) [launch adaptive-gap-20260910-115703]	75.39	—	1/1	619
Blur schedule chosen by a transfer-gap trigger [launch adaptive-gap2-20260910-121319]	77.13	—	1/1	622
Blur schedule chosen by the network (first trigger)	76.34	—	1/1	610
Blur schedule, more time at high blur	76.57	—	1/1	663
Blur schedule, more time at low blur	78.74	—	1/1	654
Resolution schedule chosen by a transfer-gap trigger	77.29	—	1/1	410

continued on next page

Configuration	Acc. (%)	SD	Seeds	Wall (s)
adaptgap hi [50000 train / 10000 eval, 120 epochs] (budget 120 epochs: not comparable on accuracy with 30-epoch arms)	86.73	—	1/1	2639
adaptgap lo [50000 train / 10000 eval, 120 epochs] (budget 120 epochs: not comparable on accuracy with 30-epoch arms)	78.40	—	1/1	2598
gplateau hi [50000 train / 10000 eval, 120 epochs] (budget 120 epochs: not comparable on accuracy with 30-epoch arms)	87.16	—	1/1	2594
gplateau lo [50000 train / 10000 eval, 120 epochs] (budget 120 epochs: not comparable on accuracy with 30-epoch arms)	80.68	—	1/1	2765
plain hi [50000 train / 10000 eval, 120 epochs] (budget 120 epochs: not comparable on accuracy with 30-epoch arms)	86.04	—	1/1	1752
plain lo [50000 train / 10000 eval, 120 epochs] (budget 120 epochs: not comparable on accuracy with 30-epoch arms)	77.10	—	1/1	1774
Full-data plain / Gaussian plateau / geometric				
Full-data plain / Gaussian plateau / geometric (fulldata_gaussian; owner Max). Conditions: reference recipe; evaluation every 2 epochs; this run produced the pinned campaign asset set. Evaluation split: test. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation..				
geometric_seed0	78.46	0.72	3/3	684
plain_seed0	75.40	0.40	3/3	458
plateau_seed0	78.41	0.32	3/3	693
Progressive input resolution, one seed				
Progressive input resolution, one seed (progressive_resolution_pilot; owner Max). Conditions: reference recipe, seed 0; pairs with fulldata_gaussian seed 0. Evaluation split: test. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation..				
progres_gauss_seed0	79.93	—	1/1	674
progres_seed0	79.57	—	1/1	452
ResNet-20 BN pilot, plain vs internal Gaussian				
ResNet-20 BN pilot, plain vs internal Gaussian (resnet20bn_gaussian_pilot; owner Max). Conditions: 10,000-image subset, 2,400 updates, val split; seed 0. Evaluation split: val. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation..				
gaussian_r20bn_seed0	59.62	—	1/1	174
plain_r20bn_seed0	55.36	—	1/1	119
ResNet-18 BN, plain vs internal Gaussian				
ResNet-18 BN, plain vs internal Gaussian (resnet18bn_gaussian_pilot; owner Max). Conditions: 10,000-image subset, val split; ResNet-18 BatchNorm; seed 0. Evaluation split: val. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation..				
gaussian_r18_seed0 [launch resnet18-gaussian-20260908-144640]	62.16	—	1/1	197
gaussian_r18_seed0 [launch resnet18-gaussian-complete]	62.16	—	1/1	197
plain_r18_seed0	54.26	—	1/1	151
db2 wavelet shrinkage pilot (thread closed)				
db2 wavelet shrinkage pilot (thread closed) (db2_pilot; owner Max). Conditions: 10,000-image subset, 2,400 updates, val split; seed 0. Evaluation split: val. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation..				
db2_r20bn_seed0	56.34	—	1/1	3856
plain_r20bn_seed0	55.40	—	1/1	89
GroupNorm LR diagnosis and 10,000-image plain vs Gaussian				
GroupNorm LR diagnosis and 10,000-image plain vs Gaussian (gn_lr_audit; owner Max). Conditions: 10,000 train images, 5,000 val; ResNet-20 GroupNorm; 1,200 updates; val accuracy, not test. Evaluation split: val. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation..				
gaussian_seed0	46.91	0.60	3/3	61
gaussian_seed0_lr0p02	47.94	—	1/1	66
plain_lr0p005	41.78	—	1/1	25
plain_lr0p02	34.74	—	1/1	26
plain_seed0	50.36	0.52	3/3	47
plain_seed0_lr0p02	50.72	—	1/1	49
First internal-filter pilots (prelim checks, one collapsed run)				
First internal-filter pilots (prelim checks, one collapsed run) (pilot_internal_gaussian; owner Max). Conditions: 10,000-image subsets, ResNet-20 GroupNorm; the first pilot collapsed to uniform prediction (LR too high). Evaluation split: val. Timing: wall_seconds of the run as recorded by continuation_driver; includes periodic evaluation..				
db2	23.34	—	1/1	648
gaussian [launch pilot-continuation-20260908-122558]	20.26	—	1/1	33
gaussian [launch pilot-manual]	20.62	—	1/1	31
plain [launch pilot-continuation-20260908-122558]	14.10	—	1/1	24
plain [launch pilot-manual]	14.12	—	1/1	23

continued on next page

Configuration	Acc. (%)	SD	Seeds	Wall (s)
Fixed Gaussian input blur, GroupNorm				
Fixed Gaussian input blur, GroupNorm (exp0_input_gaussian; owner Max). Conditions: CIFAR-10 45,000 train / 5,000 stratified val; ResNet-20 GroupNorm; 14,040 updates; fixed sigma per run. Evaluation split: val. Timing: not recorded.				
level_0	84.11	0.60	3/3	—
level_0p5	82.55	0.06	3/3	—
level_1	71.33	2.38	3/3	—
level_2	48.32	2.96	3/3	—
level_3	38.85	1.86	3/3	—
Input-blur warm starts, GroupNorm				
Input-blur warm starts, GroupNorm (exp1_input_warmstart; owner Max). Conditions: as exp0; warm-start branching from blurred prefixes. Evaluation split: val. Timing: not recorded.				
arm_P	83.19	0.51	3/3	—
arm_W	82.93	0.11	3/3	—
prefix_sigma_warm	55.57	2.10	3/3	—